Stat 266 – Unit 6 – Video Set 5 – Class Examples

Summary:

Let $Y_1,\ldots,Y_n$ be independent and identically distributed continuous random variables with cumulative distribution function $F_Y(y)$ and density function $f_Y(y)$. We denote the ordered (ranked) variables by $Y_{(1)},\ldots,Y_{(n)}$, so that $Y_{(1)} \le Y_{(2)} \le \cdots \le Y_{(n)}$.

e.g. $Y_{(1)} = minimum(Y_1,\ldots,Y_n)$ and $Y_{(n)} = maximum(Y_1,\ldots,Y_n)$.

- The density of the $k-th$ ordered (ranked) variable ($Y_{(k)}$) is given by

$$f_{Y_{(k)}}(y) = \frac{n!}{(k-1)!(n-k)!}\left[F_Y(y)\right]^{k-1}\left[1-F_Y(y)\right]^{n-k} f_Y(y),$$ support is the same as Y.

- The joint density of the $j-th$ and $k-th$ ($j<k$) ordered (ranked) variables is given by

$$f_{Y_{(j)},Y_{(k)}}(y_j,y_k) = \frac{n!}{(j-1)!(k-1-j)!(n-k)!}\left[F_Y(y_j)\right]^{j-1}$$
$$\times\left[F_Y(y_k)-F_Y(y_j)\right]^{k-1-j}\left[1-F_Y(y_k)\right]^{n-k} f_Y(y_j)f_Y(y_k),\ y_j<y_k.$$

The distribution of $X_{(n)} = \max\{x_1,\ldots,x_n\}$.

$$P(X_{(n)}<y) = P(x_1<y,\ x_2<y,\ \ldots,\ x_n<y) \overset{\text{(independent)}}{=\!=} \prod_{i=1}^{n} P(X_i<y) = P(X<y)^n.$$

Take deriviative with respect to y, $f_{Y_{(n)}}(y) = n\,P(X<y)^{n-1}\cdot f_X(y)$.

The distribution of $X_{(1)} = \min\{x_1, \cdots, x_n\}$.

$$P(x_{(1)} < y) = 1 - P(x_{(1)} > y) = 1 - P(x_1 > y, x_2 > y, \cdots, x_n > y).$$

$$= 1 - \prod_{i=1}^{n} P(x_i > y) = 1 - \prod_{i=1}^{n} (1 - P(x_i \leq y)) = 1 - (1 - P(x \leq y))^n.$$

Take deriviative on both sides, we have

$$f_{x_{(1)}}(y) = -n(1 - P(x \leq y))^{n-1} \cdot f_x(y).$$

The distribution of k-th order $Y_{(k)}$. Let $G(y) = P(Y \leq y)$.

$P(Y_{(k)} < y) = P($ At least K samples are less than $y)$

$$= \sum_{i=k}^{n} P(\text{totally } i \text{ samples are less than } y).$$

$P(\ i$ samples are less than $y) \xlongequal{\text{binomial distribution}} C_n^i (1-G(y))^{n-i} (G(y))^i$

$P(Y_{(k)} < y) = \sum_{i=k}^{n} C_n^i (1-G(y))^{n-i} G(y)^i$ Take derivative both sides,

we have $f_{Y_{(k)}}(y) = \left(G(y)^n + \sum_{i=k}^{n-1} C_n^i (1-G(y))^{n-i} G(y)^i \right)'$.

$$= \underline{n G(y)^{n-1} \cdot f_Y(y)} + \sum_{i=k}^{n-1} \left(C_n^i (n-i)(1-G(y))^{n-i-1} (-f_Y(y)) + \underline{C_n^i \cdot i (1-G(y))^{n-i} \cdot G(y)^{i-1} \cdot f_Y(y)} \right)$$

combine them

$$= -\sum_{i=k}^{n-1} \frac{n!}{(n-i-1)!\, i!} (1-G(y))^{n-i-1} \cdot f_Y(y) + \sum_{i=k}^{n} \frac{n!}{(i-1)!\,(n-i)!} (1-G(y))^{n-i} G(y)^{i-1} f_Y(y)$$

a lot of terms cancel each other, only remain is: $f_Y(y) \frac{n!}{(k-1)!\,(n-k)!} (1-G(y))^{n-k} G(y)^{k-1}$

Other sight to explain the density function of $Y_{(k)}$.

$$f_{Y(k)}(y) \approx P(Y_{(k)} \in [y, y+dy]).$$

that is to choose 1 Y_i from $Y_1, \dots, Y_n$, such that $Y_i \in [y, y+dy]$, choose $k-1$ from $n-1$, s.t they are less than y, and the rest is greater than $y+dy$.

$$P(Y_{(k)} \in [y, y+dy]) = C_n^1 \cdot C_{n-1}^{k-1} \cdot G(y)^{k-1} \cdot (1-G(y))^{n-k} \cdot P(y < Y_i < y+dy).$$

$$P(y < Y_i < y+dy) = G(y+dy) - G(y) \overset{\text{Taylor expansion}}{=} G(y) + G'(y) \cdot dy + o(dy^2) - G(y) \approx f_Y(y) \cdot dy$$

$$f_{Y(k)}(y) \approx C_n^1 C_{n-1}^{k-1} G(y)^{k-1} (1-G(y))^{n-k} \cdot f_Y(y).$$

Similiarly, one may have intuition of distribution $Y_{(j)}$ and $Y_{(k)}$.

$(j<k)$. $\quad f_{Y_{(j)}, Y_{(k)}}(y_i, y_k)$

We need $j-1$ - Ys such that Y is less than y_i,

need 1 Y such that Y is near y_i

need $k-1-j$ Ys such that Y is between y_i and y_k,

and 1 Y such that Y is near y_k.

and $n-k$ Y such that Y is greater than Y_k

$$f_{Y_{(j)}Y_{(k)}} = C_n^{j-1} C_{n-j+1}^{1} G(y_i)^{j-1} \cdot f_Y(y_i) \cdot C_{n-j}^{k-1-j} \left(G(y_k) - G(y_i)\right)^{k-1-j}$$

$$C_{n-k+1}^{1} \cdot f_Y(y_k)\,(1-G(Y_k))^{n-k}.$$

Example A : Insurance claim amounts (in thousands of dollars) following a car accident are independent and identically distributed according to the following density function

$$f_Y(y)=\begin{cases}\dfrac{2}{(y+1)^3} & , y>0\\ 0 & , otherwise\end{cases}.$$

Among a sample of n accidents, what is expected value of the smallest claim?

We know that $P(Y_{(1)}<y) = 1 - P(Y_{(1)}<y)$

$= 1 - (1-G(y))^n$

take derivative : $n(1-G(y))^{n-1} \cdot f_Y(y)$.

$$G(y) = \int_{-\infty}^{y} f_Y(y)\,dy = \begin{cases}\int_0^y \frac{2}{(y+1)^3}\,dy = 1-\frac{1}{(y+1)^2} \\ 0 \qquad y\le 0\end{cases}$$

The density of $Y_{(1)}$ is $n(y+1)^{-2(n-1)}\left(\frac{2}{(y+1)^3}\right) = 2n \cdot \frac{1}{(y+1)^{2n+1}}$.

$$E(Y(1)) = \int_0^{+\infty} y \cdot 2n \cdot \frac{1}{(y+1)^{2n+1}}\, dy$$

Let $u = y+1$ $\quad du = dy$

$$\Rightarrow (2n)\int_1^{+\infty} (u-1) \cdot \frac{1}{u^{2n+1}}\, du = (2n)\int_1^{+\infty} \frac{1}{u^{2n}} - \frac{1}{u^{2n+1}}\, du$$

$$2n\left(-\frac{1}{(2n-1)} u^{-(2n-1)} - \frac{1}{(-2n)} u^{-2n}\right)\Bigg|_2^{+\infty}$$

$$= (2n)\left(\frac{1}{2n-1} - \frac{1}{2n}\right) = \frac{1}{2n-1}$$

Example B : Consider a device that is hooked up to two batteries, where their lives are independent. Let X be the lifetime of battery 1, and Y be the lifetime of battery 2, both measured in days. Suppose $X \sim Exponential(\mu_X = \beta)$ and $Y \sim Exponential(\mu_Y = 2\beta)$. If the device will run as long as at least one battery functions, find the density function that describes the life of the machine, and the expected life of the machine.

$T = \max(X, Y)$.

$T(t) = P(T < t) = P(X < t, Y < t) = P(X < t) \cdot P(Y < t)$.

$P(X<t) = \int_0^t \frac{1}{\beta} e^{-\frac{x}{\beta}} dx = -e^{-\frac{x}{\beta}} \Big|_0^t = 1 - e^{-\frac{t}{\beta}}$

$P(Y<t) = \int_0^t \frac{1}{2\beta} e^{-\frac{x}{2\beta}} dx = -e^{-\frac{y}{2\beta}} \Big|_0^t = 1 - e^{-\frac{t}{2\beta}}$

$T(t) = (1 - e^{-\frac{t}{\beta}})(1 - e^{-\frac{t}{2\beta}})$ take derivative $f_T(t) = \frac{1}{\beta} e^{-\frac{t}{\beta}} (1 - e^{-\frac{t}{2\beta}})$

$+ \frac{1}{2\beta} e^{-\frac{t}{2\beta}} (1 - e^{-\frac{t}{\beta}})$

$$f_T(t) = \begin{cases} \frac{1}{\beta} e^{-\frac{t}{\beta}} \left(1 - e^{-\frac{t}{2\beta}}\right) + \frac{1}{2\beta} e^{-\frac{t}{2\beta}} \left(1 - e^{-\frac{t}{\beta}}\right) & t > 0 \\ \text{otherwise.} & 0 \end{cases}$$

$$E(T) = \int_0^{\infty} t\left(\frac{1}{\beta} e^{-\frac{t}{\beta}}\left(1 - e^{-\frac{t}{2\beta}}\right)\right) + t\left(\frac{1}{2\beta} e^{-\frac{t}{2\beta}}\right)$$

$$\left(1 - e^{-\frac{t}{\beta}}\right) dt$$

$$= \int_0^{+\infty} \frac{t}{\beta} e^{-\frac{t}{\beta}} dt - \int_0^{+\infty} \frac{1}{\beta} t e^{-\frac{3t}{2\beta}} dt + \int_0^{+\infty} \frac{t}{2\beta} e^{-\frac{t}{2\beta}}$$

$$- \frac{t}{2\beta} e^{-\frac{3t}{2\beta}} dt$$

$$= \beta - \frac{4}{3}\beta + 2\beta - \frac{2\beta}{3} \times \frac{1}{3}\beta = \frac{7}{3}\beta$$

Example C : The proportion of decay for two substances are jointly distributed random variables, with joint density function given by

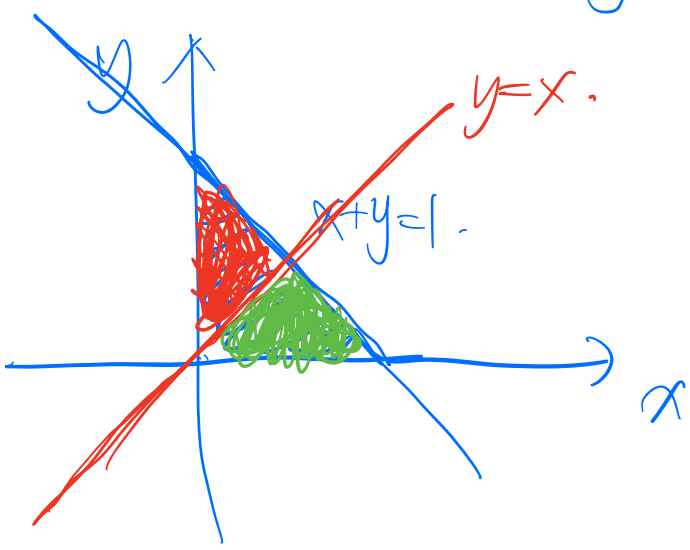
$$f(x,y)=\begin{cases}60xy^2 & , 0<x<1-y<1\\ 0 & , \text{otherwise}\end{cases}$$

Find the expected value for the substance with the lowest decay.

Let $U = \min(X, Y)$.

$$E(U) = \int_0^1 \int_0^1 \min(x,y)\, f_{X,Y}(x,y)\, dx\, dy.$$

Now consider the region of $f(x,y)$.



$0 < x < 1-y \Rightarrow$ $x+y<1$ & $0<x$

$1-y<1 \Rightarrow 0<y.$

red region, $y > x$,

Green region $y < x$.

In red region, $\min(x, y) = x$.

we have $\iint_{red} x \cdot 60xy^2 \, dx\,dy = \int_0^{\frac{1}{2}} \int_x^{1-x} x \, 60xy^2 \, dy\,dx$

$= \frac{1}{18}$

In Green region, $\min(x, y) = y$

$\iint y \cdot 60xy^2 \, dx\,dy = \int_0^{\frac{1}{2}} \int_y^{1-y} y \cdot 60x^2y^2 \, dx\,dy = 0.0125$

Green

add them together

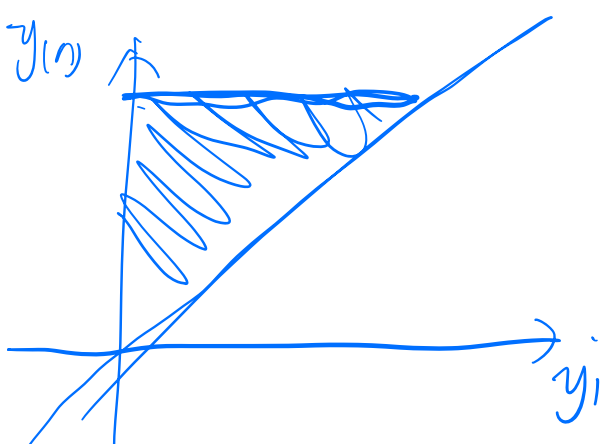
Example D : On a dangerous stretch of road of length 1, accidents occur independently at random (uniformly) over the length of the road. At one end of the road there is an ambulance dispatch centre. Suppose it is located at position "0", and the end of the road is position "1". Let Y denote the one-way travel distance to an accident from the dispatch centre. Thus, $Y \sim Uniform(0,1)$. Suppose on a particular day there were n accidents.

a) Consider the distance between the furthest accident and the closest accident to the dispatch centre, $Y_{(n)} - Y_{(1)}$. Find the probability density and expected value of $Y_{(n)} - Y_{(1)}$.

b) Consider $U = Y_{(1)} / Y_{(n)}$ and $V = Y_{(n)}$. Find $f_{U,V}(u,v)$.

The joint distribution of $Y_{(1)}$ and $Y_{(n)}$ is given by

$$g_{(1)(n)}(y_1, y_n) = \begin{cases} n(n-1)\left(F(\overset{y_n}{\overset{\shortparallel}{y_n}}) - F(\overset{y_1}{\overset{\shortparallel}{y_1}})\right)^{n-2} \cdot 0 < y_1 < y_n < 1. \\ 0 \quad \text{otherwise} \end{cases}$$



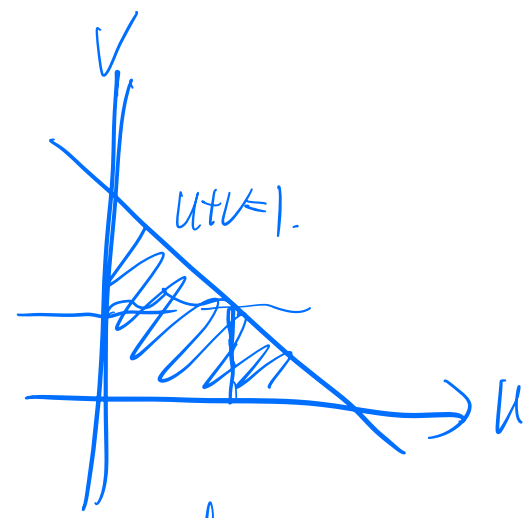
Let $U = Y_{(n)} - Y_{(1)}$ $\quad V = Y_{(1)}$.

$Y_{(n)} = U + V$ $\quad Y_{(1)} = V$. $\quad J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ ~~=0~~.

$\det J = 1$. By method of transformation, we have

$$g_{U,V}(u,v) = n(n-1)(u+v-v)^{n-2} \cdot 1 = n(n-1)u^{n-2}.$$

$$= \begin{cases} n(n-1)u^{n-2} & 0 < u+v < 1 \quad 0 < v < 1 \\ 0 & \text{otherwise} \end{cases}$$



The margin density of U is $\int_0^{1-u} n(n-1)u^{n-2}\,dv$.

$$= n(n-1)\, u^{n-2} v \Big|_0^{1-u} = n(n-1)\, u^{n-2}(1-u). \qquad 0 < u < 1$$

$$0. \qquad \text{otherwise}$$

$$E(U) = \int_0^1 u \cdot n(n-1)\, u^{n-2}(1-u)\, du$$

$$= n(n-1) \int_0^1 u^{n-1} - u^n \, du$$

$$= n(n-1) \cdot \left(\frac{u^n}{n} - \frac{u^{n+1}}{n+1} \Big|_0^1 \right) = n(n-1)\left(\frac{1}{n} - \frac{1}{n+1} \right).$$

$$= \frac{n-1}{n+1}$$

b) $U = \dfrac{Y_{(1)}}{Y_{(n)}}$ $\quad V = Y_{(n)}$ $\qquad U V = Y_{(1)}$ $\quad V = Y_{(n)}$.

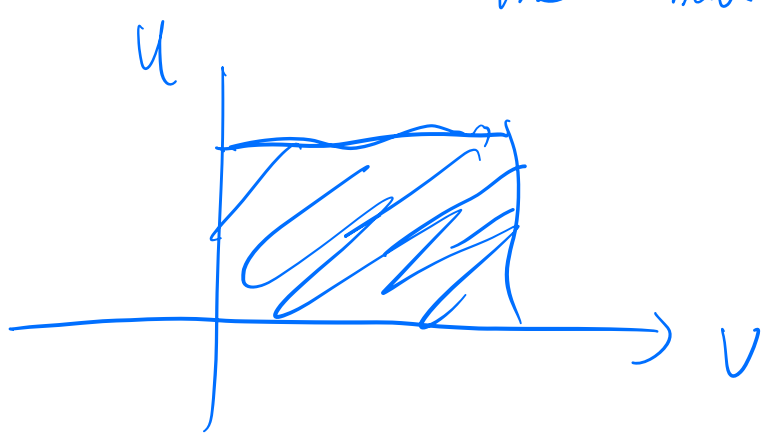
$$\text{Jacobian} = \begin{bmatrix} v & u \\ 0 & 1 \end{bmatrix} \qquad \det J = v.$$

$$g_{UV}(u,v) = n(n-1)(v-uv)^{n-2} \cdot v.$$

since $0 < v < 1$.

$uv = Y_{(1)} < Y_{(n)} = v \quad 0 < u = \frac{Y_{(1)}}{Y_{(n)}} < 1.$

we have the region is $uv < v$ & $0 < v < 1$.



$uv - v = 0$

$v(u-1) = 0.$

we got $g_{UV}(u,v) = \begin{cases} n(n-1)(v-uv)^{n-2} \cdot v & v \in [0,1] \\ & u \in [0,1] \\ 0 & \text{otherwise.} \end{cases}$